BLOCKCHAIN • SMART CONTRACTS

GradEscrow

Milestone-Based Smart Contracts with Adaptive Dispute Windows

PROJECT REVIEW 1 • PROPOSAL & DESIGN

Under the Guidance of: **Dr. Anubha S Pearline Ma'am**

Rudraksh Choudhary
Reg. No. 25BCE1790

Ayush Abhinav
Reg. No. 25BCE5359

1 Introduction: What is GradEscrow?

Decentralized Escrow

The Core Concept

A decentralized, trustless escrow smart contract built for the freelance gig economy that replaces centralized third parties with autonomous blockchain logic.

Tranche-Based Release

Milestone Payments

Instead of all-or-nothing payments, project budgets are split into verifiable milestone tranches, unlocking fair compensation as work progresses.

Trust-Decay Mechanism

Adaptive Dispute Window

Our primary innovation: dispute review windows dynamically shrink following each successfully verified milestone as intra-contract trust accumulates.

Project Vision:
To eliminate arbitrary payment delays and excessive intermediary commissions in freelance contracting by mathematically modeling earned trust directly into smart contract settlement logic.

1 Problem Statement: Flaws in Traditional Escrow

1

All-or-Nothing Release

Traditional platforms lock 100% of project funds until final completion. If a dispute or scope change occurs near the end, freelancers risk losing pay for weeks of honest intermediate work.

2

Expensive Centralized Intermed

Platforms like Upwork and Fiverr charge steep 10% to 20% commission rakes, plus withdrawal fees, while relying on slow, subjective manual dispute resolution taking 14 to 45 business days.

3

Static, Unchanging Dispute Win

Every milestone is subjected to the same fixed 7-to-14 day review hold. An honest contractor with 5 perfect deliverables is treated identically to an unverified first-time actor.

Core Problem Identified:
In existing freelance escrow protocols, honest contractors suffer cumulative weeks of idle capital lockup because dispute windows remain static regardless of earned reliability.

1 Motivation: Why This Matters Now

High Platform Commissions

Current Reality:
Freelance platforms extract up to 20% of contractor gross earnings. For small vendors and gig workers, this is a massive financial drain.

Proposed Solution:
GradEscrow replaces platform fees with autonomous blockchain code, reducing overhead to pure network execution costs.

Slow Dispute Resolution

Current Reality:
Conventional support arbitration takes weeks, freezing working capital and forcing freelancers into cash-flow emergencies.

Proposed Solution:
Smart contracts enforce deterministic on-chain review windows, auto-releasing payouts upon deadline expiry.

Academic Research Gap

Current Reality:
Prior blockchain escrow papers studied milestone payouts or multi-party voting, but none modeled adaptive dispute timing.

Proposed Solution:
GradEscrow bridges this theoretical and practical gap by demonstrating mathematically decaying dispute windows.

2 Literature Survey: Prior Work & Analysis

Paper / System	Key Contribution	Identified Academic Limitation
Smart Contract Escrow (IJCRT, 2023)	Milestone-based fund release via contributor voting	Static dispute rules; no mechanism for trust decay.
Crowdfunding Milestones (2020)	Voting-based tranche release for backer fairness	Binary release system; lacks proportional milestone logic.
Construction Smart Contracts (MDPI, 2023)	Milestone tracking + automated payment on-chain	No adaptive dispute window; requires external inspectors.
E-Commerce Escrow Survey (2025)	Comprehensive survey of decentralized escrow protocols	Broad survey; lacks intra-contract mathematical adaptation.

Literature Conclusion & Synthesis:

While previous academic systems successfully demonstrate milestone-based fund releases, none mathematically model intra-contract reputation to shrink dispute latency dynamically.

3 Research Gap & Proposed Innovation

THE IDENTIFIED RESEARCH GAP:

"A milestone-escrow smart contract that adapts its own dispute-resolution timing based on accumulated verified trust within the same contract does not currently exist in academic literature."

Limitations in Existing Systems

- Dispute windows remain fixed constants regardless of project progress.
- Zero-trust is assumed for every milestone, penalizing reliable freelancers.
- Voting-based systems require external crowds/DAOs, impractical for 2-party contracts.

Our Proposed Innovation

- Incorporate mathematical trust decay into smart contract state transitions.
- Compress dispute windows safely as verified milestones succeed.
- Enforce an inviolable 24-hour safety floor (W_min) to prevent zero-day fraud.

4 Proposed Solution & Project Objectives

Objective 1: Smart Contract Design

Design an EVM-compatible Solidity smart contract architecture implementing milestone state management, role authorization, and pull-over-push fund security.

Objective 2: Adaptive Mathematical Model

Formulate and implement a bounded trust-decay equation that programmatically computes dispute durations based on consecutive on-time milestone approvals.

Objective 3: Web3 Dashboard UI

Develop an intuitive single-page web dashboard allowing clients to fund escrows and freelancers to submit deliverables and track real-time countdown timers.

Objective 4: Mobile Wallet Integration

Implement native MetaMask mobile deep-linking to ensure seamless onboarding for mobile-only freelancers without requiring desktop access.

5

Proposed Methodology & Adaptive Trust Math

Proposed Mathematical Trust-Decay Function

$$W(n) = \max(W_{min}, W_0 \times (1 - k \times n)) \quad \text{or} \quad W(n) = W_{min} + (W_0 - W_{min}) \times e^{(-\lambda n)}$$

Model Parameters

- **n:** Count of successfully verified milestones (n ≥ 0)
- **W_0:** Initial baseline dispute window (7.0 days / 168 hours)
- **k / λ:** Decay constant (e.g. k = 0.20 linear; λ = 0.35 exponential)
- **W_min:** Mandatory safety floor (1.0 day / 24 hours) to prevent zero-day attacks

Planned Milestone Progression (W_0 = 7d)

- **Milestone 1: 7.0 Days (168h)** (Baseline zero-trust window)
- **Milestone 2: 5.6 Days (134h)** (20% window reduction after M1 approval)
- **Milestone 3: 4.2 Days (101h)** (40% reduction; earned trust demonstrated)
- **Milestone 4: 2.8 Days (67h)** (60% reduction; expedited settlement)
- **Milestone 5: 1.4 Days (34h)** (80% reduction; safety floor guaranteed)

6 Proposed System Architecture & Workflow

Stage 1

Escrow Deposit

Client deposits 100% of project budget into smart contract escrow vault.

Stage 2

Milestones Locked

Milestone descriptions, tranches, and payout percentages locked on-chain.

Stage 3

Proof Submission

Freelancer finishes milestone deliverable and uploads proof URI/hash.

Stage 4

Adaptive Window

Contract starts dispute timer W(n), computed from verified milestone count.

Stage 5

Automated Release

If no dispute raised within window, contract auto-releases tranche payout.

Stage 6

Trust Feedback

Verified milestone increments trust score; subsequent dispute window shrinks.

Dispute Fallback: If client disputes deliverable within window → contract pauses payout and triggers arbitration.

7 Resources Needed & Technology Stack

Smart Contract Layer

- Language: Solidity (^0.8.20)
- Framework: Hardhat development environment
- Security: OpenZeppelin ReentrancyGuard & Ownable
- Network: Ethereum Sepolia Testnet (EVM)

Frontend & UI Layer

- Library: React 19 + Vite tooling
- Styling: Tailwind CSS responsive utility framework
- Icons & UI: Lucide React icon package
- Architecture: Dual Client & Freelancer viewports

Web3 Integration & Wallet

- Web3 Provider: Ethers.js (v6) for contract calls
- Wallet: MetaMask browser extension
- Mobile Support: Native MetaMask Mobile deep linking
- RPC: Infura / Alchemy Sepolia RPC endpoints

Testing & Verification Tools

- Unit Testing: Mocha & Chai automated test suite
- Local Blockchain: Hardhat Network with time simulation
- Code Quality: Slither static analyzer
- Version Control: Git & GitHub repository

8 Project Timeline & Plan of Work

Phase 1: Weeks 1–2

Architecture & Smart Contra

- Finalize mathematical trust-decay formulation
- Write core Solidity smart contract (GradEscrow.sol)
- Set up Hardhat environment and compile contracts

Phase 2: Weeks 3–4

Automated Testing & Securit

- Write comprehensive unit tests in Mocha/Chai
- Test time simulation (evm_increaseTime) for decay
- Validate edge cases (reentrancy, safety floor limits)

Phase 3: Weeks 5–6

Frontend & Web3 Integratio

- Build React 19 dashboard for clients & freelancers
- Integrate Ethers.js v6 with MetaMask wallet
- Implement mobile deep-linking for smartphone users

Phase 4: Weeks 7–8

Testnet Deployment & Evalua

- Deploy contract to Ethereum Sepolia Testnet
- Conduct empirical benchmarking on freelance dataset
- Prepare documentation, final demo, and report

9 Expected Outcomes & Deliverables

First Adaptive Dispute Escrow

A validated mathematical model proving that dispute windows can safely decay based on intra-contract reputation while preserving full client review protection.

Full-Stack Working DApp

An interactive React 19 web application connected to Ethereum testnet via MetaMask, enabling one-click milestone creation, funding, and auto-release.

Quantifiable Capital Lockup Reduction

Targeting a >60% reduction in average escrow turnaround latency across multi-milestone freelance engagements compared to static 7-day or 14-day models.

Open-Source & Auditable Codebase

Complete public GitHub repository containing verified Solidity contracts, automated unit test suites, frontend components, and comprehensive documentation.

BLOCKCHAIN • SMART CONTRACTS

Thank You!

GradEscrow: Milestone-Based Smart Contracts with Adaptive Dispute Windows

Under the Guidance of: **Dr. Anubha S Pearline Ma'am**

Rudraksh Choudhary (25BCE1790) • Ayush Abhinav (25BCE5359)

Project Proposal: Review 1 — Concept, Design & Methodology

Open for Questions & Panel Suggestions